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Time taken 2 mins 11 secs

Grade 10.00 out of 10.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

Which layer in a CNN is mainly responsible for extracting features from images?

- ☐ a. D) Softmax Layer
- ☒ b. C) Convolutional Layer ✓
- ☐ c. B) Fully Connected Layer
- ☐ d. A) Pooling Layer

Your answer is correct.

The correct answer is:

C) Convolutional Layer

Question 2

Correct

Mark 1.00 out of 1.00

What is the main purpose of Dropout in [deep learning](#)?

- ☒ a. B) Avoid overfitting ✓
- ☐ b. D) Speed up testing only
- ☐ c. C) Reduce batch size
- ☐ d. A) Increase image size

Your answer is correct.

The correct answer is:

B) Avoid overfitting

Question 3

Correct

Mark 1.00 out of 1.00

In Dropout with probability 0.8, what does it mean?

- ☐ a. D) All neurons are active
- ☐ b. A) 80% neurons are always removed
- ☒ c. C) 80% neurons remain active during iteration ✓
- ☐ d. B) 20% neurons are trained only once

Your answer is correct.

The correct answer is:

C) 80% neurons remain active during iteration

Question 4

Correct

Mark 1.00 out of 1.00

Which optimizer combines Momentum and RMSProp ideas together?

- ☐ a. A) SGD
- ☐ b. B) AdaGrad
- ☒ c. C) Adam ✓
- ☐ d. D) Nesterov

Your answer is correct.

The correct answer is:

C) Adam

Question 5

Correct

Mark 1.00 out of 1.00

What does the Pooling layer mainly do in CNN?

- ☐ a. C) Add new neurons
- ☒ b. B) Down-sample feature maps ✓
- ☐ c. A) Increase parameters
- ☐ d. D) Normalize outputs

Your answer is correct.

The correct answer is:

B) Down-sample feature maps

Question 6

Correct

Mark 1.00 out of 1.00

In Mini-Batch Gradient Descent, the batch size is:

- ☐ a. D) Zero
- ☐ b. A) Equal to dataset size
- ☐ c. B) Always 1
- ☒ d. C) Between 1 and dataset size ✓

Your answer is correct.

The correct answer is:

C) Between 1 and dataset size

Question 7

Correct

Mark 1.00 out of 1.00

Which optimizer measures the gradient at $\theta + \beta m$ instead of θ ?

- ☒ a. D) Nesterov Momentum ✓
- ☐ b. C) Momentum
- ☐ c. B) Adam
- ☐ d. A) RMSProp

Your answer is correct.

The correct answer is:

D) Nesterov Momentum

Question 8

Correct

Mark 1.00 out of 1.00

What is one advantage of Batch Normalization?

- ☐ a. D) Decreases learning completely
- ☒ b. B) Normalizes layer activations ✓
- ☐ c. A) Increases overfitting
- ☐ d. C) Removes convolution layers

Your answer is correct.

The correct answer is:

B) Normalizes layer activations

Question 9

Correct

Mark 1.00 out of 1.00

What is an Epoch in neural networks?

- ☐ a. B) One batch processing
- ☐ b. D) One optimizer step only
- ☐ c. A) One neuron update only
- ☒ d. C) Complete forward and backward pass of the whole dataset once ✓

Your answer is correct.

The correct answer is:

C) Complete forward and backward pass of the whole dataset once

Question 10

Correct

Mark 1.00 out of 1.00

Which activation function is commonly used after convolution layers in CNNs?

- ☐ a. B) Sigmoid
- ☒ b. C) ReLU ✓
- ☐ c. D) Tanh
- ☐ d. A) Softmax

Your answer is correct.

The correct answer is:

C) ReLU